AstroGrass: a curated resource for the Brachypodium distachyon ISS transcriptome study OSD-375 and gravitropism candidate genes

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-- DRAFT MANUSCRIPT --

Abstract

Few plant spaceflight experiments have used monocots, although monocots include most of the crops grown on Earth[Su2023]. NASA Open Science Data Repository (OSDR) study OSD-375 profiled the roots and shoots of three Brachypodium distachyon accessions (Bd21, Bd21-3 and Gaz8) grown on the International Space Station, with ground controls run at Kennedy Space Center[Su2023]. This draft describes AstroGrass, a curated resource built around that study. It contains (1) the OSD-375 sample design and growth conditions, taken from the OSDR metadata and the primary paper; (2) a catalogue of five grass spaceflight and related omics studies, each checked against its OSDR, GEO or PRIDE record; (3) 29 gravitropism candidate genes defined by Arabidopsis anchor genes, whose identifiers were checked in UniProt (12 were corrected) and mapped to Brachypodium, rice and wheat orthologs with Ensembl Compara; (4) the candidates' spaceflight differential-expression calls, taken from the per-accession tables Su et al. published; and (5) time-lapse videos of Brachypodium root gravitropism. No new analysis of the OSD-375 sequencing data is reported, and every expression value is Su et al.'s. Thirteen of the 29 candidates have at least one Brachypodium gene in those tables, seven of them through an Ensembl Compara ortholog of the anchor. Measuring gravitropism across accessions and comparing with Arabidopsis remain planned.

Keywords: *Brachypodium distachyon, gravitropism, spaceflight, microgravity, NASA OSDR, OSD-375, APEX-06, ISS, transcriptomics, monocot, data resource*

Introduction

Plants sense the direction of gravity mainly through the movement of starch-filled plastids (amyloplasts) that act as statoliths in specialised gravity-sensing cells (statocytes)[Morita2010]. In Arabidopsis, LAZY1-LIKE (LZY) proteins associate with amyloplasts and the adjacent plasma membrane and become polarised according to the direction of gravity[Nishimura2023,Chen2023]. At the plasma membrane, LZY recruits RCC1-like domain (RLD) proteins, which are required for polar auxin transport[Furutani2020], and in the root columella the auxin efflux regulator PIN3 relocalises laterally after gravity stimulation[Friml2002].

In rice, root growth angle is controlled in part by DEEPER ROOTING 1 (DRO1) and its homolog qSOR1[Uga2013,Kitomi2020]. Crops such as wheat have been studied for decades as components of life support systems for space[Wheeler2017], yet few plant spaceflight experiments have used monocots[Su2023].

Brachypodium distachyon is a model for temperate grasses. It was the first member of the Pooideae, the grass subfamily that includes wheat and barley, to have its genome sequenced[IBI2010], and de novo assemblies of 54 lines define its pan-genome[Gordon2017].

In the APEX-06 experiment (NASA OSDR study OSD-375), seedlings of the Bd21, Bd21-3 and Gaz8 accessions were germinated and grown on the International Space Station (ISS) and in a ground control[Su2023]. Su et al. reported that

the roots of all three accessions grew more slowly and had longer root hairs in microgravity, and that the number of differentially expressed genes (DEGs) differed widely between accessions: 4,527 in Gaz8, 1,353 in Bd21 and 570 in Bd21-3, with only 47 shoot DEGs and six root DEGs shared between accessions[Su2023].

This draft describes the AstroGrass resource as it now stands: what is in it, where each item comes from, and what remains to be done.

Status of this draft

Versions of this manuscript and its website before 14 October 2026 reported a 15-accession gravitropism survey, a genome-wide association study (GWAS), a re-analysis of OSD-375 with gene-level fold changes, a cross-species comparison with Arabidopsis, and synteny, promoter-motif and alternative-splicing analyses. These results have been withdrawn. The gravitropism measurements had been generated by the pipeline's demonstration mode from seeded random numbers, no genotype data or association test existed, and the remaining values were typed into the code rather than computed. The Brachypodium gene identifiers given for the candidate genes were also withdrawn, because none could be confirmed as an ortholog of its Arabidopsis anchor. The sections below report only what the repository now contains.

Results

OSD-375 study design

OSD-375 contains 48 RNA-seq samples: three accessions, two organs (roots and shoots), flight or ground control, and four APEX Growth Units per accession and condition (Figure 1). Each of the 24 growth units gave one root and one shoot sample. Seedlings were germinated for 24 h under red light and then grown for four days under continuous red, blue and green light at about 40-55 µmol m?^2 s?^1 and 22 degC; the flight samples received an absorbed radiation dose of 1.397 mGy (OSDR metadata;[Su2023]). The ground control was run in a VEGGIE unit inside the ISS Environmental Simulator at Kennedy Space Center with a two-day delay[Su2023]. Supplementary Figure S1 summarises the hardware, timeline and conditions.

All 48 libraries were sequenced on Illumina NovaSeq 6000 instruments, but Bd21 was sequenced at the University of Wisconsin-Madison with 2 x 150 bp reads, and Bd21-3 and Gaz8 by the NASA GeneLab Sample Processing Lab with 2 x 100 bp reads[Su2023]. Accession is therefore confounded with sequencing batch. Flight-versus-ground comparisons within an accession are not affected, but differences between accessions in the number of DEGs cannot be separated from batch effects.

Study catalogue

AstroGrass catalogues five studies (Figure 2): OSD-375[Su2023]; OSD-622, a microarray study of leaves of dwarf wheat (cultivar USU Apogee) grown on the ISS in the Biomass Production System during the PESTO experiment[Stutte2006]; two ground RNA-seq studies of Brachypodium, GSE97940 (roots treated with 2,4-D to induce nodule-like structures)[Thomas2018] and GSE48040 (cold-treated seedlings of accession ABR5)[Zhang2013]; and the leaf phosphoproteome PXD000868[Lv2014]. Each entry was checked against its OSDR, GEO or PRIDE record, and each DOI against CrossRef or DataCite. Fields that a record does not state are marked as such rather than filled in.

Gravitropism candidate genes

The candidate list contains 29 genes chosen from the Arabidopsis gravitropism literature: PIN and AUX/LAX auxin carriers, LAZY/DRO and SGR genes, TIR1/AFB auxin receptors, ARF and Aux/IAA transcription factors,

starch-synthesis genes, calcium and mechanosensing genes, and cell-wall genes. Each candidate is defined by an Arabidopsis anchor gene. In the list as first compiled, 12 of the 29 anchor identifiers (AGIs) named a different gene; for example, the AGI given for LAZY1 was that of PDF1B, and the AGI given for MSL10 was that of MSL1. Each was replaced by the AGI that UniProt assigns to the intended gene name (for SGR2, the gravitropism phospholipase rather than STAY-GREEN 2, which shares the name), and all 29 anchors now match their UniProt gene names (data/genotypes/arabidopsis_anchor_check.csv).

Ensembl Compara (plants, Ensembl Genomes release 63) lists Brachypodium orthologs for the anchors of 21 candidates: five one-to-one, eight one-to-many and eight many-to-many (data/genotypes/gravitropism_candidate_genes.csv). Several candidates share orthologs: the PIN3, PIN4 and PIN7 anchors map to the same two Brachypodium genes, AUX1 and LAX1 to the same three, and ARF7 and ARF19 to the same one. For eight anchors (LAX2, LAX3, LAZY1, DRO1, CRK28, CML24, EXPA1 and CSLD1) Compara lists no Brachypodium, rice or wheat ortholog, and these remain TODO. Rice and wheat orthologs from the same source are given in the table.

The BRADI identifiers given in the first version of the list were withdrawn: nine of the 29 do not exist in Ensembl, and none is a Compara ortholog of its corrected anchor (data/genotypes/withdrawn_bradi_id_check.csv). Two markers in the earlier table, a serine hydroxymethyltransferase and the chloroplast-encoded RuBisCO large subunit, were not gravitropism candidates and were dropped with the typed-in expression values attached to them.

Candidate genes in the published spaceflight DEG lists

Su et al. list their DEGs per accession and organ in Supplemental Tables S2-S4[Su2023]: 325 root and 1,027 shoot DEGs in Bd21, 224 and 345 in Bd21-3, and 475 and 4,186 in Gaz8 (Figure 3). These are counts of table rows; the totals in their abstract (1,353, 570 and 4,527 DEGs) differ slightly, and the tables are used as published. Every listed gene has a DESeq adjusted p-value and an edgeR q-value below 0.05. Their gene identifiers are Phytozome v3.1 locus names. All 5,403 distinct identifiers exist in Ensembl under the corresponding BRADI v3 names, and for a random sample of 100 genes Su et al.'s best-hit Arabidopsis ortholog was among the gene's Compara orthologs for 52, against none when the best hits were shuffled between the sampled genes (data/su2023/id_mapping_check.csv).

Thirteen of the 29 candidates have at least one Brachypodium gene in these lists (Figure 4; tables/candidate_spaceflight_degs.csv). For seven candidates (PIN2, AUX1, LAX1, IAA14, PGM, ADG1 and CAS) the gene is a Compara ortholog of the anchor: six genes in all, most of them DEGs in Gaz8. For the other six (PIN7, LAX2, DRO1, ARF7, CML24 and EXPA1) the match is only through Su et al.'s best-hit Arabidopsis ortholog; four of these anchors (LAX2, DRO1, CML24 and EXPA1) have no Compara ortholog. PIN2 appears both ways: its Compara ortholog BRADI_1g31530v3 is a Gaz8 DEG in roots (log2 fold change -0.78) and shoots (+2.05), while BRADI_2g52640v3, which Su et al. annotate with PIN2 as best hit, is lower in spaceflight in Bd21 and Bd21-3 roots. No gene matched to LAZY1 or CPK28 by either route is listed.

These values have three limits. A gene missing from the lists may not have been expressed or may have fallen short of significance, so absence is not evidence of no change. Differences between accessions are confounded with sequencing batch. And being a spaceflight DEG does not by itself link a gene to gravitropism.

Time-lapse videos and kinetics plots

The website hosts time-lapse videos of Brachypodium roots of accessions Per1 and BdTR7a after a 90 deg rotation, from the author's CyVerse Data Store. The repository also holds per-accession plots of root orientation and root-tip displacement over time for 45 accessions, added to the repository as Barker 2016 data. The numbers behind these plots are not in the repository, so no statistics are drawn from them here.

Planned analyses (not yet run)

TODO: Find Brachypodium orthologs for the eight anchors without one in Ensembl Compara (for example from gene-family trees), and decide which ortholog each candidate name refers to where Compara gives several (for example PIN1a and PIN1b).

TODO: Values for genes that are not in Su et al.'s DEG lists need a new differential-expression analysis of OSD-375, starting from the FASTQ files (the only expression data in OSDR for this study) and accounting for the confounding of accession with sequencing batch.

TODO: Add gravitropism measurements for the accession panel, with their source documented. The kinetics plots above would need their underlying numbers before any comparison between accessions.

TODO: An association study would need genotype data for the phenotyped accessions; with tens of accessions it would be a candidate-gene comparison rather than a GWAS.

TODO: Compare Brachypodium spaceflight responses with the author's Arabidopsis spaceflight analysis, which is in preparation, once its consensus gene set can be regenerated from code.

Methods

OSD-375 metadata

The OSD-375 file listing and ISA-Tab metadata were downloaded from the OSDR API (https://osdr.nasa.gov/osdr/data/osd/files/375) by code/01_download_osdr_osd375.py. The sample table (s_GLDS-375.txt) was parsed by code/02_parse_osdr_metadata.py to count samples per accession, organ and condition (Figure 1). Growth conditions and the radiation dose were read from the same table; the timeline and sequencing details are from Su et al.[Su2023].

Study catalogue

Catalogue entries (code/10_build_astrograss_db.py) were checked on 14 October 2026 against the OSDR study metadata API, NCBI GEO (E-utilities and the GEO sample records), the PRIDE Archive API, CrossRef and DataCite.

Candidate-gene anchors and orthologs

For each AGI in the curated list (data/genotypes/candidate_genes_curated.csv), and for each AGI it replaced, code/14_check_arabidopsis_anchors.py retrieved the reviewed UniProt entry cross-referenced to that Araport identifier (UniProt release 2026_03). An anchor was accepted only if the intended Arabidopsis gene symbol was among the entry's gene names. Replacement AGIs were chosen by hand from a UniProt search for the exact Arabidopsis gene name, and were then checked by the same script. code/15_map_orthologs.py retrieved the Brachypodium, rice and wheat orthologs of each accepted anchor from Ensembl Compara (plants) through the Ensembl REST API (Ensembl Genomes release 63), retrying server errors, and checked each withdrawn BRADI identifier against Ensembl and against those orthologs. code/03_candidate_genes.py combines the list, the anchor check and the orthologs into data/genotypes/gravitropism_candidate_genes.csv.

Spaceflight DEG lists

Supplemental Tables S2-S4 of Su et al.[Su2023] (CC BY 4.0) were taken from the article's supplementary archive through Europe PMC; data/su2023/SOURCE.md records the source and SHA-256 hashes. code/16_su2023_degs.py reads the three tables, converts each Phytozome v3.1 locus name to its Ensembl identifier (Bradi3g01250 to BRADI_3g01250v3), counts DEGs by accession, organ and direction (sign of the DESeq log2 fold change), and lists the rows for genes that are either Compara orthologs of a candidate's anchor or annotated by Su et al. with the anchor as

best-hit Arabidopsis ortholog. With -check-ids it checks the identifier conversion against Ensembl as described in Results.

Data and code availability

All code and tables are available at https://github.com/dr-richard-barker/brachypodium-gwas-spaceflight, and the website at https://dr-richard-barker.github.io/brachypodium-gwas-spaceflight/. Raw sequencing data are available from NASA OSDR under accession OSD-375 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-375; DOI: 10.26030/2x6b-3v89).

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Author Contributions

R.B. conceived the study, curated the resource, built the data pipeline and website, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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